

Jinam Shah

jinams47@gmail.com | github.com/jinamshah | linkedin.com/in/jinamshah

Engineer with **5+ years** of expertise designing and deploying scalable **ML** solutions in **NLP**, **Computer Vision**, and **Big Data**. Proven track record building robust, production-grade pipelines, optimizing **distributed training** workflows, and improving system performance through effective **MLOps** strategies. Skilled in leveraging **cloud** and **HPC** resources to deliver impactful AI systems. Adept at **problem-solving**, **innovation**, and translating **research** to practical applications.

Work Experience

Data Scientist II, LexisNexis | June 2025 - Present

- Spearheading the migration of entity recognition and resolution service infrastructure from **TorchServe** to **NVIDIA Triton Inference Server** to optimize model serving throughput and reduce latency.
- Leading the system architecture design and model serving framework, improving product reliability and reducing production errors to near-zero.
- Collaborated with enterprise cross-functional teams to design and deploy an enterprise-wide **Deep Learning platform** enabling scaling across NLP teams.

Software Engineer, Plant Sciences Initiative - North Carolina State University | July 2023 - June 2025

- Engineered a scalable data processing pipeline to process the largest agricultural imagery dataset (**300K+ bot-captured images**), automating transfer, storage, and annotation.
- Built plant detection model based on **YOLOv11** for identifying crops, weeds, and calibration objects; customized bounding box handling for **Ultralytics** implementation.
- Automated iterative model retraining loop with YOLOv11 object detection, streamlining labeled data integration, enhancing annotation pipeline throughput by **4x**, and reducing manual efforts significantly through robust **MLOps** practices.
- Architected **HPC-based** image processing workflow (**Slurm, Singularity**), saving **\$100K+** annually vs cloud.
- Developed an end-to-end **drone imagery processing pipeline** (handling **200GB+ per flight**) with a **full-stack** geospatial analytics interface (**Flask, Node.js**), reducing research overhead by **70%** and improving productivity.

Machine Learning Intern, Cactus Communications Inc | May 2022 – Nov 2022

- Constructed and optimized **production ML pipeline** for researcher disambiguation using **AllenAI's S2AND**, achieving **95% accuracy** across **250TB+** of academic data.
- Implemented **distributed** inference pipeline on AWS EMR, reducing processing time by **60%**.
- Contributed critical scalability improvements to the **open-source S2AND** repo.

Senior Software Engineer (Data & ML), Cactus Communications Inc | June 2018 - July 2021

- Engineered **BERT-based (LLM)** ensemble achieving **2-min processing** (from 8hrs) for **1500-class** classification. Built training and inference pipeline and interactive frontend; saved **\$1M annually** via serverless architecture handling **100K+ daily** requests.
- Built **serverless ML inference pipeline** handling **100K+ daily requests** with **sub-second** latency, achieving **99.99%** uptime through auto-scaling.
- Deployed **ML/NLP platform** handling **1M+ papers/month** with inference cost savings of **60%**.
- Architected **large-scale ETL pipelines** and **serverless ML infrastructure** to process over **250TB** of academic documents and 8TB of weekly data, achieving **80%** improvement in throughput and **70%** reduction in operational costs.
- Created a **distributed** data processing system using **24K CPU cores** and **48TB RAM** to generate **4.5TB** in **2.5 hours** at **1/5th** the cost of proposed cost.
- Led collaboration with **AWS S3 team** to resolve critical throughput bug affecting **500K+ daily transactions**, improving system stability by **85%**.
- Optimized multi-cloud infrastructure (**AWS/GCP/Azure**), reducing spend by **40%** and increasing performance **3x**.

Notable Projects

Master's Thesis: Author Name Disambiguation | PyTorch, transformers, Hugging Face, pandas

- Formulated **LLM embedding-based** approach achieving **70% accuracy** without demographic features across **1M+ papers**.

Automated Plant Detection Pipeline | YOLOv11, PyTorch, Ultralytics, HPC, Slurm

- Trained **object detection model** for plant/weeds recognition using **300K+** annotated field images. **Iteratively retrained model** with new labeled data. Achieved **high precision** in field detection and reduced manual labeling efforts drastically.

Bias Detection in Text | PyTorch, transformers, NumPy, scikit-learn

- Realized **92% bias reduction** while maintaining model performance within **3%** through novel vector normalization approach.

Image Caption Generator | PyTorch, CNN, LSTM, Flickr-30K

- Implemented attention-based adversarial network achieving **96% accuracy** on Flickr-30K dataset.

Explainable Grammar Correction | PyTorch, Horovod, NLTK, spaCy, AWS EC2

- Developed **distributed LLM** training pipeline processing **100K+ papers monthly**; achieved **85% accuracy** in explanation generation.

Ethical Image Compliance System | AWS Lambda, scikit-learn, OpenCV, S3

- Built serverless CV pipeline achieving **99.8% accuracy** across **50K+ monthly** image validations with **sub-second** latency.

Education

Master of Science in Computer Science, North Carolina State University | GPA: 4.0

Coursework: High-Performance Machine Learning, Neural Networks, Natural Language Processing, Algorithms for Data-Guided Business Intelligence

Thesis: Author Name Disambiguation in Academic Records using Large Language Models

Bachelor of Technology in Computer Science, NMIMS University | GPA: 3.51

Coursework: Artificial Intelligence, Predictive Modeling, Data Warehousing and Management, Image Processing, Data Analytics, Software Engineering

Research: Image Recognition and Image Caption Generation

Technical Skills

- ML & AI:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers, BERT, Hugging Face, YOLOv11, CNN, RNN, LSTM, Attention Models, Neural Networks, Deep Learning, Computer Vision, NLP, Reinforcement Learning, Generative AI, Distributed Training, GPU Acceleration, TPU utilization, Model Explainability (XAI), Hyperparameter Optimization, Transfer Learning, MLOps
- Big Data & Cloud:** Apache Spark, PySpark, Hadoop, Hive, AWS (EC2, S3, Lambda, ECS, EMR, Redshift, API Gateway, SageMaker), GCP (TPUs), Airflow, Kubernetes, ETL Pipelines, Data Lakes, Data Warehousing, Feature Engineering, Geospatial Analytics, NumPy, Pandas, OpenCV, NLTK, spaCy, Matplotlib, Seaborn, Data Visualization, SQL

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- **Programming & Development:** Python, SQL, Flask, Django, React.js, JavaScript, REST APIs, GraphQL, Microservices, Docker, Kubernetes, Singularity, Git, Linux, Shell Scripting, HPC, Slurm, CI/CD, Jenkins, GitHub Actions
- **Development Practices:** Distributed Systems, System Design, Serverless Architecture, Cloud Architecture, Infrastructure as Code (IaC), Scalability, Performance Optimization, Agile Methodology, Scrum, Technical Leadership, Code Reviews, Continuous Integration & Deployment